GTIIT Spring 2025 Calculus 1M
Workshop 12

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint or a big clue— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

Exercise 1. Determine whether the sequence converges or diverges. If it converges, find the limit:

(i) $a_n = \frac{4^n}{1+9^n}$

(ii) $a_n = \cos\left(\frac{n\pi}{n+1}\right)$

Exercise 2. For what values of r is the sequence $a_n = nr^n$ convergent?

Exercise 3. (i) If $\{a_n\}$ is convergent, show that

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} a_{n+1}$$

(ii) A sequence $\{a_n\}$ is defined by $a_1 = 1$ and $a_{n+1} = 1/(1+a_n)$ for $n \geq 1$. Assuming that a_n is convergent, find the limit of the sequence.

Exercise 4. Find the values of x for which the series converges and compute the sum of the series:

- $\sum_{n=1}^{\infty} (-5)^n x^n$
- $\sum_{n=0}^{\infty} \frac{\sin^n x}{3^n}$

Exercise 5. Determine whether the series is convergent or divergent:

- $\sum_{n=2}^{\infty} \frac{\ln(n)}{n^2}$
- $\sum_{n=1}^{\infty} ne^{-n^2}$

Exercise 6. Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms and $\sum b_n$ is convergent. Prove that if

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 0,$$

then $\sum a_n$ is also convergent.

Exercise 7. Use previous exercise to prove that the series converges:

- $\sum_{n=1}^{\infty} \frac{\ln n}{n^3}$
- $\sum_{n=1}^{\infty} \frac{\ln n}{\sqrt{n} e^n}$